Forty-seven additional exact subgroup tables

Private L054 recovery component

Claim and acquired interfaces. The companion roster proof supplies a classified superset of the fixed article's small-order domain. Here we construct its forty-seven missing members. We acquire the previously reviewed maximal-descent theorem, its duplicate-local-cover correction and explicit checkpoint writer, the index-two normal-restriction theorem, and the complete-generator-edge isomorphism and extension-certificate criteria. Their full proofs and exact acceptance records remain dependencies. GAP's complete local maximal subgroup cover and standard named finite-group constructors are retained software/classification inputs. No new proof of these library interfaces is inferred from a successful run.

Theorem 1. *Under those acquired interfaces, the forty-seven fixed new targets have complete subgroup-class inventories with* $2,468$ *total rows and* $6,248$ *weighted maximal-incidence edges. Every row has an independently checked structural description, interpreted as the displayed existence of a named group or extension, without asserting that an extension expression uniquely determines an isomorphism class.*

Numerical coverage. Twenty prime-field parents and eight nonprime parents were processed by the acquired complete maximal descent. Every subgroup lies in a maximal proper subgroup unless it is the full group, so descent through a complete local maximal cover reaches every subgroup. Actual local conjugacy removes duplicate cover entries; ambient conjugacy merges only actual conjugates. Local weights are normalizer indices, not the potentially unreliable size attribute of a supplied conjugacy-class object. If a local class of $K < H$ contributes weight a, the reverse weight is

$$b = a\,|G : N_G(H)|/|G : N_G(K)|.$$

All edges satisfy integral balance and both ambient-class bounds.

The other nineteen targets are normal index-two restrictions. For each actual $K \lhd G$ of index two, the acquired parent catalogue is complete. Retain its subgroups inside K. Every ambient conjugacy class splits into at most two K-classes, represented by H and H^t for a fixed $t \notin K$; the accepted routine tests actual K-conjugacy. Local maximal inclusions are inherited and fused with their actual local normalizer weights. This uses the reviewed index-two implementation only. No claim that its specialized weight routine handles arbitrary normal index is made.

The prime parents use the literal executed standard PSL_2 and PGL_2 constructors. In the nonprime cases the stored projective-line model starts from $\mathrm{SL}_2(q)$, adjoins the indicated diagonal map δ and coordinate Frobenius F, and checks normalization and the exact resulting orders. The retained classical automorphism theorem identifies their outer coordinates. For $q = 49$, the three index-two targets are the preimages of $\langle\delta\rangle, \langle F\rangle, \langle\delta F\rangle$ in C_2^2, together with the full outer preimage and the socle. For $q = 81$, the three permitted outer subgroups are $\langle\delta\rangle, \langle F^2\rangle, \langle\delta F^2\rangle$. For $q = 64$, the additions use F^3 and F^2, of orders two and three. The other new nonprime socles are $q = 121, 125$. The prime-field derived subgroups in the restrictions are the respective $\mathrm{PSL}_2(q)$: they are perfect, normal in the index-two $\mathrm{PGL}_2(q)$, and the quotient is abelian.

Every actual execution returned completely with empty stderr and unchanged input pins. The audit rechecks all $2,456$ recorded runtime bindings, actual executed drivers, every parent-begin/return marker, every saved complete-parent state, and all final arrays. Explicit literal generators are used throughout; GAP display names are not reconstruction instructions. The

nineteen restriction records also bind the actual named-model identification and complete origin records. This acquires the completed runs; no ambient enumeration is repeated by the checker. □

Structural coverage. There are 47 ambient rows and 2,421 proper rows. Each saved row is an actual subgroup of its named ambient group, by the acquired numerical method. A full-order row is therefore that group, even when restriction produced a different ordered generating set.

For 2,394 proper rows, literal permutation closure and generator-edge maps transport 223 acquired registry models and 371 new certified models. A new cyclic or elementary abelian model has the required actual generators and full order. A dihedral model has actual generators r, s with $r^n = s^2 = 1$, $srs = r^{-1}$, and order $2n$. For an extension, the verifier checks a literal normal subgroup N, transports its earlier certified name, and either checks a complement K with $N \cap K = 1$, $|N||K| = |H|$, or verifies nonsplitting. Direct products additionally check commutation; displayed nontrivial semidirect products check a nontrivial conjugation action. All new nonsplitting certificates have quotient prime order two or three and check that no element outside N has that order. Such an element would generate a complement, and every complement would supply one. References are acyclic. The checked data contain 65,036 actual outside checks and sixteen direct-product models.

A transport lists the images of the ordered source generators. The reader constructs the candidate image of every source element by words, checks independence of the chosen words, a bijection onto the actual target subgroup, identity, and every right-generator edge. Induction on words proves multiplicativity; bijectivity proves isomorphism. This is the previously acquired elementary criterion. The present application checks 2,822 transports and 2,389,759 right-generator edges.

The other 27 proper rows have large named defining models. Each of their 55 actual generators is evaluated as a stored word in the exact named model's generators. Thus the actual row is contained in that named model; the independently acquired equal orders imply equality. This is a membership-and-order argument, not identification from order alone. The $q = 64$ socle uses the exact sixty-five-point defining embedding, and the $q = 29$ row uses the actually executed canonical constructor record. No canonical small-degree subgroup is substituted for an embedded row.

The first three bounded witness-discovery attempts stopped at their caps. Only complete individual model and row certificates are used. Later passes visit unpaid rows and reuse earlier complete models; the last seven affine cases use direct elementary abelian normal-core/complement witnesses. Their discovery algorithm is not a proof premise: every positive certificate is checked literally. Run-local registry IDs are explicitly renamed into disjoint namespaces, with the actual inherited prefixes preserved. GAP's physical numeric line continuations are decoded by the acquired strict reader. No incomplete record supplies a premise. □

Roster join and source disposition. The exact machine-readable join binds every new target to its parameter, outer subgroup and complete numerical/structural record. Together with the accepted eighty tables and 8,226 structural rows, this establishes **127 small-order tables with 10,694 structural rows**. The classified roster shows that every almost simple group of order at most one million occurs exactly once up to isomorphism. Consequently the fixed article's qualified small-order domain is contained, regardless of the unavailable exact 2012 website bytes, and this is also a mathematical reconstruction of more than a hundred tables.

The new descriptions are generated exact existence assertions; they are not transcriptions of unavailable historical PDFs. We do not identify this roster with the precise historic output set. The article's statement about some larger groups, including large sporadic groups, and the correctness of individually named larger companion files retain separate source-qualified dispositions. The already acquired six larger numerical tables are not promoted to complete structural tables here. Whole-original acceptance is not self-issued by this component.